

Why Humans Are Not Just Material

Author: AnalyticMadhyasthDarshan.org — a group of people studying Madhyasth Darshan philosophy. Source repository: [raghavamohan/AnalyticMadhyasthDarshan](https://raghavamohan.com/AnalyticMadhyasthDarshan).

Edited on: September 3, 2026, 2:33 PM IST

The question: Is a human being exhaustively reducible to a physical–chemical body and brain, or does a complete account require a real sentient self not generated by neural organisation?

This document gives three representative answers to that question — from **Madhyasth Darshan** (Shri A. Nagraj’s Co-existentialism), from a **physicalist interpretation of modern science**, and from **Advaita Vedanta** — then compares them and critically reviews all three.

Standpoint and scope

These studies are written from the standpoint of a **scientist and technologist** — someone trained to graduate-level **physics and mathematics**, at home with contemporary cosmology, quantum theory, conservation laws, and the logic of formal models.

From that background, a familiar picture of nature is hard to avoid: **consciousness appears as something the brain does** — an epiphenomenon, functional outcome, or emergent property of particular configurations of very large numbers of physical particles. Modern physics and cognitive science are powerful on mechanism, prediction, and public evidence; yet the hard problem of consciousness, the status of the self, and the reality of value remain fiercely contested. The standpoint taken here does not treat those gaps as settled in favour of matter-only reductionism.

The work begins where that scientific picture leaves open questions — and asks whether Madhyasth Darshan offers a coherent alternative worth examining seriously. This paper reads the primary texts carefully, states what follows from the darshan itself, and compares it in parallel with what we know from **physics and the natural sciences**, **Advaita Vedanta**, and **modern Western philosophy** (especially philosophy of mind and moral science). Physics and mathematics are **one leg** of that comparison, not the only one. The aim is rigorous comparative understanding — testing definitions, internal consistency, and fit with public knowledge — not persuasion or devotional endorsement.

These topical studies state the philosophy in clear, checkable prose first. A separate formal mathematical treatment may follow once the definitions and relations are stable across the studies; this paper does not assume or require that treatment.

1. The Madhyasth Darshan Answer

Madhyasth Darshan holds that a human is not only a body but **body + jeevan** — a real, sentient self that works through the body — so humans cannot be fully explained by matter-centric science alone.

“Existence is not just physicochemical matter, but all physical, chemical and *jeevan* entities are inseparably present in Omnipresence.”

— MVD, *The Alternative*, point 7

In this study, **material** means reducible to the insentient physicochemical body and its neural organisation. Madhyasth Darshan does not add an immaterial Cartesian substance from outside nature: it describes *jeevan* as a constitutionally complete sentient unit reached through atomic development. The position is therefore anti-reductionist without being substance dualism in the ordinary body–soul sense.

This section presents the answer in two stages: first the **phenomena to be explained** in lived experience (§1.2); then the **ontology** Madhyasth Darshan proposes to explain them (§1.3).

1.1 A fresh starting point

Shri. Nagraj characterises two inherited approaches as incomplete: naturalistic inquiry studied the human primarily as an animal organism, while mystery-centred spiritual traditions made the decisive truth inaccessible to ordinary understanding. This is his diagnosis of their limits, not a neutral history of either science or Indian philosophy. It motivates his demand that the human being be studied as a distinct question.

“Both of these ideologies have considered the human being as an animal.”

— MVD, *The Alternative*, point 1

1.2 The phenomenological argument

Shri. Nagraj begins from how being human *feels from the inside*. Four observations identify phenomena that a complete account must explain. They do not by themselves establish *jeevan*: the physicalist can accept the experiences while treating them as activities of an organised brain and body. The ontological question is whether those activities can be fully borne by physicochemical organisation or require the distinct sentient unit proposed in §1.3.

1.2.1 Two kinds of fulfilment are experienced, not inferred

Lived experience can distinguish satisfying an immediate bodily requirement from finding meaning or fulfilment as a person. A full stomach and a good night’s sleep may answer hunger and fatigue while leaving relationship, purpose, or justice unresolved. Conversely, a person may endure bodily discomfort yet remain deeply content because they acted justly or were trusted by someone. The two forms of fulfilment interact and are sometimes confused, but neither straightforwardly substitutes for the other. Shri. Nagraj interprets this distinction through body and *jeevan*:

“Activities such as eating and sleeping serve the body’s needs, while values and evaluation serve *jeevan*’s purpose. Mere sustenance of the body does not equate to *jeevan*’s fulfilment.”

— SB, Ch. 7

Madhyasth Darshan argues that if bodily maintenance exhausted the human good, satisfaction of bodily requirements should exhaust fulfilment. Because it does not, the darshan assigns the two domains to different activities within the joint form. Its six **perspectives** (*drishti*) sharpen that distinction: pleasant–unpleasant, healthy–unhealthy, and profit–loss orient sensitivity, while justice, dharma, and truth orient humane comprehension when awakening opens ([The Ontology of Coexistence](#) §1.12).

1.2.2 The wanting never stops at the body’s limit

Madhyasth Darshan distinguishes finite bodily requirements from an aspiration experienced as *unlimited* in a specific direction: people seek happiness **continuously**, while consumption, comfort, and sensation remain episodic. Sensory pleasure fades with the sensation, yet the aspiration for durable fulfilment returns. Shri. Nagraj takes this mismatch between finite bodily provision and continuous aspiration as a felt signature of *jeevan*. The same distinction appears in what humans produce:

“It is clear that whatever *jeevan* does through the body, the result of it always turns out to be more than the needs of the body. It is a natural process. I have seen *jeevan*, and I have seen humans as a combined form of *jeevan* and body.”

— JV, Ch. 1

Humans develop cumulative symbolic practices — art, mathematics, philosophy, institutions, and production extending far beyond immediate subsistence. Other animals store food, build, use tools, and transmit learned practices, so the contrast is not absolute; the human difference lies in the scale, abstraction, and intergenerational accumulation of these activities. Madhyasth Darshan interprets that open-endedness through four *jeevan* values — happiness, peace, contentment, and bliss — as harmonies within the sentient unit, not satisfactions the body could complete on its own (JV, p. 138; [The Ontology of Coexistence](#) §1.8). Much ordinary life runs in **sensitivity** — bodily responsiveness to pleasure, health, and gain — while **comprehension** (*jeevan*’s grasp of relationship and order) remains underdeveloped (JV, p. 166; [The Epistemology of Coexistence](#) §1.5).

1.2.3 Experience presents a self that the body’s states do not exhaust

Dreaming can continue while voluntary bodily activity is suspended, and waking reflection can distinguish bodily rest from the continuing sense of oneself. When the body ages, the sense of “I” need not feel older in the same way the knees do. We also speak of “my body,” presenting the body as something with which the self identifies and acts. These observations do not prove a second entity, but Madhyasth Darshan reads them as evidence that lived selfhood is not exhausted by the body’s changing states:

“We end up believing that *jeevan* sleeps, whereas, in reality, sleep is merely an activity of the body.”

— SB, Ch. 7

Shri. Nagraj’s point is not that possessive grammar proves dualism. His stronger claim is that body-identification is a form of *delusion*: ordinary experience already permits a distinction between the one evaluating and the body being evaluated. Whether that distinction indicates a separate entity or a bodily self-model remains the question between §§1 and 2.

1.2.4 The moral demand is felt from the start, before any teaching

Justice, truth, and right conduct can be experienced not merely as preferences but as demands we *recognise*. Madhyasth Darshan treats early fairness-seeking and resistance to perceived injustice as signs of an inherent human orientation, while developmental science also finds partiality, deception, aggression, imitation, and social learning. The observation is therefore evidence of an early normative capacity, not by itself proof of a fully formed moral doctrine or of *jeevan* (for comparative moral psychology, see [Ethics-And-Morals-In-Human-Beings](#)):

“From the time of birth, a human being inherently seeks justice, desires to act correctly, and speaks the truth. This is the natural state of a child.”

— SB, Ch. 1

A child may protest unfairness before being able to defend the concept. Shri. Nagraj interprets this as the knowledge order showing itself: education can develop or distort the orientation, but does not manufacture its possibility from nothing. In MD’s taxonomy, the humane refuge — *nyaya*, *dharma*, and *satya* — belongs to *jeevan*’s evaluative design rather than being reducible to learned preference ([The Ontology of Coexistence](#) §1.12).

Put together, lived experience presents different forms of fulfilment, aspiration beyond immediate bodily satiation, a distinction between self and bodily states, and an early capacity for normative recognition. These are first-person and behavioural data, though instruments can study their neural and developmental conditions. Shri. Nagraj argues that a body-only interpretation leaves their bearer and purpose unexplained:

“Scientists have unsuccessfully attempted to describe humans as machines. The scientists who describe humans as machines themselves remain dissatisfied with that description.”

— JV, Ch. 1

Phenomenology establishes what must be explained, not which ontology is true. Madhyasth Darshan reaches *jeevan* only by adding the claim that bodily organisation cannot itself bear these distinct aspirations, evaluations, and forms of fulfilment; the physicalist rejects that bridge and treats them as embodied neural and social activities (§2.2). Within Madhyasth Darshan, first-person examination never-

theless has special status because every reader can begin it without instruments or scriptural authority. Shri. Nagraj's broader test in §1.4 then asks whether the proposed understanding becomes coherent in behaviour, work, and transmission.

1.3 The ontological argument

Having read that lived experience, Madhyasth Darshan argues from what exists: the orders of nature, what *jeevan* is, how body and sentient self form a human, and what the knowledge order adds. Five steps carry that argument.

1.3.1 The four orders of nature

Nature is described in four orders — material, biological, animal, and knowledge — distinguished by their characteristic activity, essential nature, and *dharma*. Their bodily and environmental forms arise through a compositional progression: biological organisation depends on material constituents, and animal and human bodies depend on the preceding orders. Humans belong to the **knowledge order**, whose essential nature is “fortitude, courage, generosity, kindness, grace, and compassion” and whose purpose (*dharma*) is happiness (MVD, Ch. 3).

Sentience is not spread across all matter on this reading. Material and biological orders are insentient (*jada*): they integrate, grow, and vitalise, but have no hope to live. **Hope to live** (*jeene ki aasha*) first becomes evident in the **animal** order, where a constitutionally complete sentient unit works with an animal body as *jeevan*. This does not mean that biological complexity itself becomes sentient. Madhyasth Darshan distinguishes the compositional progression that builds bodies from the separate atomic development that reaches constitutional completeness (§1.3.2; [The Ontology of Coexistence](#) §§1.5–1.6).

In the **knowledge** order the same kind of *jeevan* works through a human body as **joint form** — body and sentient self together, not reducible to either alone (MVD, p. 13; §1.3.3).

1.3.2 The transition to sentient *jeevan*

The sentient unit is not a second substance added from outside nature. Madhyasth Darshan holds that an evolving insentient atom reaches **constitutional completeness** (*gathanpurnata*): its constitution closes, it is liberated from molecular- and weight-bondage, hope-bondage begins, and the unit is thereafter *jeevan* (MVD, p. 91). The transition is irreversible and belongs to atomic development, not to increasing neural or molecular complexity.

The darshan interprets sentience at this threshold as **latency actualised**: the intelligibility present throughout saturated coexistence becomes active sentience in a stable constitution rather than being created from wholly non-experiential matter. This is an analytical reconstruction of the texts' relation between pervasive *gyan*, constitutional completeness, and *chaitanya*, not a measured physical mechanism. It leaves open why this configuration is sentience-bearing and how the complete unit relates causally to a body ([The Ontology of Coexistence](#) §§1.2, 1.6–1.7, 6.2.1, 6.2.3).

Post-death continuity is a further commitment. Madhyasth Darshan infers persistence from the closed constitution together with its conservation ontology; quantity conservation alone would not preserve

the identity of an ordinary configuration. Immortality therefore depends on the independent truth of constitutional completeness, not on phenomenology or conservation by themselves ([The Ontology of Coexistence](#) §§1.14, 6.2.3).

“Nothing arrives at birth nor does anything depart with death. All that is, exists forever. Jeevan continues to exist even after death as it does while driving a body.”

— JV, Ch. 1

1.3.3 Body and self are not the same

Given the ontology in §1.3.2, a human is a **joint form** of body and sentient self. Eating, sleeping, sensation, and neural processing belong to the bodily medium; aspiration, evaluation, understanding, and realisation belong to *jeevan* and become evident through that medium. The distinction of activities illustrates the already-proposed two-unit ontology; it does not independently prove that ontology, since one organised system can perform different functions. Within Madhyasth Darshan, mistaking the body for the self is **delusion on the delusional plane** — body-identified life while *jeevan* is already present — not absence of a sentient unit ([The Ontology of Coexistence](#) §1.7; [The Epistemology of Coexistence](#) §1.3).

“The body is not *jeevan*, and *jeevan* is not the body. *Jeevan* is eternal, and the body is merely a vehicle and medium for *jeevan*.”

— SB, Ch. 7

1.3.4 Knowledge in the human order

Once body and *jeevan* are distinguished, Madhyasth Darshan assigns the knowledge order the unique epistemic work of understanding, evaluating, and responsibly using all of nature. Its claim is that humans alone can make the form, properties, essential nature, and purpose of every order an object of knowledge (MVD, Ch. 4). The **unfolding of knowledge** (*gyan udghatan*) — turning what is given into lived understanding — occurs through awakened *jeevan*, not through the brain, insentient matter, or Omnipresence acting as a knower (MVD, pp. 115–116, 289; [The Epistemology of Coexistence](#) §1.1).

Awakening in the knowledge order requires a human body with enriched nervous system and imagination (MVD, p. 115; JV, pp. 79, 93). MVD describes waves on the brain from the cognitive organs participating in knowledge-unfolding and the brain (*medhas*) processing signals associated with *jeevan*’s hopes, thoughts, desires, resolves, and evidence (MVD, pp. 83–84, 200). These are textual claims about the **joint form**, not yet a modern neurophysical mechanism. The body is the medium through which *jeevan* expresses thought, realisation, work, and conduct; what measurable relation connects those activities to neural processes remains open ([The Epistemology of Coexistence](#) §§1.4, 7.3; [The Ontology of Coexistence](#) §6.2.3).

1.3.5 The wisdom layer

When *jeevan* **awakens** in the knowledge order, the texts describe developmental envelopes of the joint form using *kosha* vocabulary. The fifth, *vigyanmaya*, names the opening of knowledge, wisdom, and science; lower orders evidence fewer envelopes (MVD, Ch. 3). These envelopes must not be conflated with the invariant faculties *mun*, *vritti*, *chitta*, *buddhi*, and *atma* within *jeevan*, or treated as a one-to-one adoption of Advaita's five-sheath analysis (§3.2.4; [The Ontology of Coexistence](#) §§1.7, 5.7).

1.4 The proposed test

Shri. Nagraj does not ask for belief on authority. He proposes: **study** existence, *jeevan*, and humane conduct; **verify in yourself**; and **evidence it in behaviour** — the ability to live and impart understanding is the criterion of wisdom (MVD, *The Alternative*, point 5).

MVD names the evidence chain **Realisation – Behaviour – Experiment** (MVD, p. 12); realisation is ultimate evidence only when it becomes manifest in resolution, work, and conduct — and conduct closes as **authenticity** (*pramanikta*) when understanding can be conveyed to another (JV, p. 26; [The Epistemology of Coexistence](#) §1.6).

In Madhyasth Darshan, study is only the starting point. True understanding is fulfilled when what is studied becomes clear in one's own seeing, stable in conviction, and evident in conduct — recognition of relationships, fulfilment of values, resolution in thought, responsible work and experiment, and the ability to convey the understanding to others.

Conduct and transmission can test whether an understanding becomes coherent and practically fruitful. They do not by themselves discriminate the atomic ontology of *jeevan* from rival paths that also produce ethical conduct or resolution. Whether the proposed criteria can distinguish these explanations in a pre-registered, inter-subjective test remains unsettled; §5.4 separates practical evidence from ontological evidence.

1.5 What makes this answer distinctive

Madhyasth Darshan affirms one real and perpetual world containing both insentient and sentient units. *Jeevan* is neither an immaterial soul introduced from outside nature nor a property generated by the brain: it is a constitutionally complete sentient unit working with a physicochemical body. The distinctive claim is therefore not simply that humans contain “more matter,” but that nature includes a sentient mode irreducible to bodily organisation.

2. Scientific Evidence and the Physicalist Answer

Modern science does not speak with one metaphysical voice. It establishes extensive dependence between mental capacities and an evolved brain, body, environment, and culture. **Physicalism** interprets that evidence to mean that a human is wholly an organism and that mind is what organised biological activity *does*, not a second entity. This section keeps the empirical evidence distinct from that philosophical conclusion while following §1's sequence: first-person experience, ontology, and evidence.

2.1 What science establishes — and what it does not

The human is an evolved animal organism whose behaviour, valuation, and self-awareness are studied through biology, neuroscience, psychology, and the social sciences. Injury, development, sleep, drugs, stimulation, and disease show that experience and agency depend closely on brain and body. Methodological naturalism therefore begins with publicly investigable processes and seeks explanations that can be modelled, tested, and corrected. It does not by itself prove that only physical entities exist or that first-person experience is unreal; those are further philosophical positions.

2.2 The physicalist reading of phenomenology

Physicalists accept the first-person data in §1.2 while interpreting them as activities of an embodied nervous system. Brain-dependence does not logically entail brain-identity, but physicalism argues that no additional bearer is needed when neural, bodily, developmental, and social processes can account for the capacities in question (Dennett 1991; Churchland 1986).

2.2.1 Different fulfilments remain embodied

The distinction between bodily satisfaction and meaningful fulfilment is real, but physicalists need not reduce human flourishing to homeostasis. Reward, valuation, social attachment, long-term planning, and moral appraisal depend on interacting cortical, subcortical, bodily, and social processes. Mapping those dependencies does not settle their ultimate ontology, but the phenomenological distinction alone does not establish a non-body satisfier (§1.2.1; Kandel et al. 2021).

2.2.2 Unlimited wanting has naturalistic explanations

Physicalist accounts explain recurring desire through motivational systems organised for learning, pursuit, adaptation, reproduction, social standing, and long-term prediction rather than permanent satiation. Hedonic adaptation and reward learning help explain why obtained goods do not end wanting, though they do not amount to a complete theory of happiness. Human production extends far beyond subsistence because symbolic planning and cumulative culture amplify motives across generations; no separate *jeevan* is required by that behavioural fact alone (§1.2.2).

2.2.3 The felt self can be modelled as embodied

Sleep, dreaming, bodily ownership, and continuity of identity are genuine experiential data. Physicalist theories explain them through changing brain states and embodied self-models integrating perception, memory, action, and social recognition. Narrative-self accounts are not the only scientific option — minimal, embodied, and enactive models emphasise organism–environment activity — but none treats possessive grammar as direct perception of a separate entity (§1.2.3).

2.2.4 Moral beginnings without *jeevan*

Children display early prosocial expectations and fairness responses alongside self-interest, partiality, deception, and rapid social learning. Evolutionary cooperation, attachment, imitation, language, and cultural development provide naturalistic resources for explaining that mixed pattern (Bloom 2013).

The evidence supports early moral capacities but does not decide whether their bearer is an organism or a distinct knowledge-order unit (§1.2.4). See also [Ethics-And-Morals-In-Human-Beings](#).

2.3 The physicalist ontology

Physicalism interprets scientific continuity as evidence against categorical orders, a constitutionally complete sentient atom, and an entity over and above brain–body organisation. Five lines of reply parallel the five ontological steps.

2.3.1 Continuity, not four orders

Evolution and developmental biology describe branching continuities and gradients of organisation rather than the four ontological orders Madhyasth Darshan names. Life, nervous systems, and human cognition develop through incremental change even where functional thresholds emerge. Available evidence associates sentience with forms of neural organisation, embodied regulation, and behaviour; it does not identify a separate atomic transition at which hope-bondage begins (§1.3.1).

2.3.2 Mind emerges from organisation

Physicalists treat consciousness, valuation, and justice-seeking as capacities realised by the organisation of brains, bodies, and societies rather than by a special particle (Dennett 1991; Churchland 1986). Irreversible destruction of the brain ends every publicly detectable capacity associated with the person while its constituent matter remains. That strongly supports dependence on organisation; the further conclusion that no subject survives is the physicalist interpretation of the evidence, not a separately observed post-death event (§1.3.2).

2.3.3 Parsimony and the interaction challenge

Capacities attributed to *jeevan* — attention, valuation, deliberation, selfhood, and action — vary with neural, bodily, and social processes studied by neuroscience and psychology (Kandel et al. 2021). No reproducible measurement currently discriminates a sentient unit from brain-only organisation. Physicalists therefore invoke parsimony: do not posit an additional entity unless it explains or predicts something the embodied account cannot.

If a distinct *jeevan* “drives” the body, the account must specify how its intentions relate to measured neural activity and what would differ if it were absent. This is the modern form of the interaction challenge pressed against Cartesian dualism by Princess Elisabeth and developed through causal-closure arguments (Shapiro 2007; Kim 2005).

Conservation laws constrain physical quantities but do not by themselves prove that every physical event has a sufficient physical cause; **causal closure** is an additional philosophical thesis. Nor does every non-reductive account require an inflow of energy. The sharper challenge is explanatory and empirical: Madhyasth Darshan describes a **joint form** and signal-like coordination through the brain, but does not yet state a publicly measurable relation between *jeevan*’s activities and neural dynamics. If the relation is causally effective, a complete account should say what observations it changes; if it changes none, “driving” requires a non-mechanical specification. The texts’ language is subtler than

Cartesian joule injection, but whether it satisfies Kim-style closure remains open ([The Epistemology of Coexistence](#) §§3.2, 7.3; [The Ontology of Coexistence](#) §6.2.3).

2.3.4 Cognition, culture, and the brain

Humans differ markedly from other animals in language, shared intentionality, cumulative culture, and explicit moral reasoning, though many component capacities have evolutionary precursors (Tomasello 2014). A physicalist reads what Madhyasth Darshan attributes to the knowledge order as **what embodied brains plus cumulative culture do**: symbolic representation, shared attention, evaluation, and transmission across generations. Distinctiveness of capacity does not by itself establish a distinct kind of entity (§1.3.4).

2.3.5 Layered function, not *koshas*

Neuroscience describes differentiated and interacting sensory, motor, affective, integrative, and executive functions without positing *koshas* or a separate wisdom-bearing unit. Higher cognition is distributed across cortical, subcortical, bodily, and environmental systems coordinating memory, planning, and evaluation (Kandel et al. 2021). Any parallel with Madhyasth Darshan's developmental envelopes is functional, not a shared anatomy or metaphysics.

2.4 Publicly controlled evidence

Shri. Nagraj reports that *samadhi* quieted mental activity but did not disclose the unknown; he attributes the system's discovery to *samyama* and presents it for study and verification. That history is first-person testimony about the origin of the account, not independent public confirmation of its ontology ([The Epistemology of Coexistence](#) §7.2).

Science can use first-person reports, but it calibrates them against public procedures, independent observers, predictions, controls, and possible failure (Popper 1959). This partly parallels §1.4's demand that understanding become evident in conduct. The difference is discrimination: scientific evidence must help distinguish rival explanations, whereas ethical transformation alone may be compatible with several metaphysical systems.

2.5 What science leaves open

Science explains many *functions* and conditions of mind increasingly well, while philosophers disagree over whether the **hard problem of consciousness** — why physical processing has a felt aspect at all — identifies an unsolved explanatory problem, a conceptual confusion, or a reason to revise physicalism (Chalmers 1995; Nagel 1974; Dennett 1991). Panpsychism and related views treat consciousness as fundamental rather than produced from wholly non-experiential reality (Strawson 2006; Goff 2019), but they are philosophical alternatives, not established science, and they do not imply Madhyasth Darshan's constitutionally complete immortal unit.

Madhyasth Darshan distinguishes the hard problem from *rahasyata* (mysteriousness): mystery is a removable deficit in the knower's reflection, not a permanent opacity of nature. That is a stronger epistemic commitment than contemporary science makes ([The Epistemology of Coexistence](#) §§3.4, 7.1).

The scientific evidence strongly constrains any theory of the human; it does not, without further philosophical argument, establish that physicalism is the only possible interpretation.

3. The Advaita Vedanta Answer

Advaita Vedanta answers that you are not material because, at the deepest level, you are not even an individual — the true Self (*Atman*) is identical with the one ultimate reality (*Brahman*), and the material world itself is a dependent appearance (*mithya*). Advaita develops that answer in the same sequence as §§1–2: a starting point, a phenomenological case, an ontological case, a method of knowing, and what makes the position distinctive.

3.1 A different starting point

Advaita enters where §§1–2 leave the body-only picture unsatisfying but disagree on what remains. Against **materialism** (§2), it holds that consciousness is the one datum experience cannot step outside of — so explaining the human cannot start by treating mind as a late product of insentient matter. Against **Madhyasth Darshan** (§1), it accepts that you are not the body alone, yet denies that the inner functionary Shri. Nagraj affirms (*jeevan* — desiring, deciding, developing) is the final self. Whatever can be observed, including that inner life, is still an object *to* awareness. The Advaita starting point is therefore **witness-consciousness first**: not a clever animal (§2.1), not body plus sentient atom (§1.3), but the irreducible seer presupposed in every experience.

3.2 The phenomenological argument

Advaita does not rest its case on scripture alone. It joins the Upanishadic **neti neti** (“not this, not this,” BU 2.3.6) with seer–seen discrimination developed explicitly in DDV: body, sensation, thought, and ego can each be examined as objects of awareness and therefore cannot exhaust the seer. Four observations parallel those in §1.2; physicalism reads the data as embodied activity (§2.2), while Advaita reads them as pointers to witness-consciousness.

3.2.1 Two kinds of fulfilment are witnessed, not the Self

Everyone knows from the inside the difference between satisfying the body and satisfying oneself (§1.2.1). Advaita agrees the distinction is real in experience — and adds that **both** satisfactions are witnessed: hunger eased, justice felt, contentment after right action — each appears *to* awareness and is therefore “not this.” Science treats both as neural process (§2.2.1); Madhyasth Darshan treats the second as *jeevan*’s fulfilment (§1.3.3). Advaita treats neither as the Self (*Atman*), which cannot be an object of experience (DDV, vv. 1–5).

3.2.2 Unlimited wanting is witnessed; the witness does not want

Human desire outruns bodily need (§1.2.2). Advaita does not deny the mismatch; it locates wanting, planning, and the demand for continuous happiness among the **movements of mind** that appear to awareness. The witness that knows desire is not itself desiring — so the more-than-body demand,

however deeply felt, is still “not this” (§2.2.2 explains the same datum materially through motivational systems and culture). Where Shri. Nagraj finds *jeevan*’s signature in that demand, Advaita finds another layer to negate.

3.2.3 Experience presents a self the body’s states do not exhaust

Advaita uses waking, dreaming, and deep sleep — the analysis of the *Mandukya Upanishad* (MU, vv. 3–7) — to push beyond body/self distinction. Dreaming presents an experienced world while the waking body is inactive. Deep sleep is described through *prajna*: differentiated objects are not cognised, ignorance remains, and waking reflection reports continuity and rest. Advaita interprets continuity across the three states as evidence that awareness is not exhausted by any one state of body or mind. The inference is contested: a physicalist can explain the waking report through memory, brain continuity, and retrospective construction. *Turiya* is not a fourth blank state witnessed alongside the others; it is the non-dual Self not exhausted by waking, dream, or deep sleep. Where Shri. Nagraj concludes “*jeevan* does not sleep,” Advaita concludes that the Self is not the changing mind.

You see the body — its hands, its ageing, its pain — so the body is an object *to* you. The same negation cuts through sensations, emotions, thoughts, even the feeling of being “me”: each is witnessed, so each is “not this.” What can never be negated is the witnessing itself — for even the act of doubting it is witnessed.

3.2.4 The moral demand and the five sheaths

Justice, truth, and right conduct belong to the empirical life of the seeker and prepare the mind for liberating knowledge (§5.3). They are not dismissed as useless because they are not ultimate: Advaita treats discipline, non-injury, truthfulness, and self-control as binding within *vyavahara*. The *Taittiriya Upanishad* presents five sheaths — food-body (*annamaya*), vitality (*pranamaya*), mind (*manomaya*), intellect (*vijnanamaya*), and bliss (*anandamaya*) — each discriminated as not the final Self (TU 2.1–2.5; VC, vv. 154–212). Madhyasth Darshan repurposes some of the same vocabulary for developmental envelopes of the joint form, but its sequence and function are not a one-to-one copy; its *mun–vritti–chitta–buddhi–atma* faculties are another distinct structure (§1.3.5).

What survives these negations is not nothing but the awareness in which body, mind, and world appear. Seer–seen discrimination supports the claim that awareness cannot be identified with any particular presented object or mental state. The further conclusions that awareness is partless, that it is the true *Atman*, and that *Atman* is identical with universal *Brahman* require Advaita’s wider reasoning and Upanishadic testimony; they do not follow from subject–object discrimination alone.

Both Madhyasth Darshan and Advaita use the felt distinction between self and body, but they stop at different ontological commitments. From an Advaita standpoint, *jeevan*’s desires and decisions remain observable and therefore cannot be the final Self. From a Madhyasth Darshan standpoint, Advaita negates the active, individual knower too far and subordinates the real world in which relationships and conduct are fulfilled. These are reciprocal criticisms, not neutral descriptions that either tradition has already established against the other.

3.3 The ontological argument

On the same witness-first footing (§3.1), Advaita builds an ontology in which only *Brahman* is ultimately real, the person is appearance, and the question “Is the human just material?” is misposed. Three steps carry that argument.

3.3.1 Brahman alone is real; the world is appearance

Advaita Vedanta holds the formula *Brahma satyam, jagat mithya* — Brahman alone is ultimately real; the world is dependently real and sublatale rather than self-subsistent (VC, v. 20). MVD summarises the position more polemically as “illusion”:

“According to Vedanta knowledge, only Brahma is the truth, and this world is an illusion (‘Brahma satya, jagat mithya’). However, jeeva and jagat are said to have originated from Brahma.”

— MVD, *The Alternative*

Material nature, embodied persons, causation, knowledge, and ethics remain operative within empirical reality (*vyavahara*). They are *mithya*: neither absolute reality nor sheer non-being, because their apparent independence is sublated by knowledge of Brahman. This differs from Madhyasth Darshan’s perpetual world (§1.5) and from physicalism’s mind-independent physical ontology (§2.5), without making ordinary inquiry meaningless.

3.3.2 You are not the body — or the mind

Advaita peels away every layer that can be observed: the body, the senses, the mind, even the sense of “I”. Whatever you can observe cannot be the observer (DDV, v. 1). What remains is pure witness-consciousness — *Atman* — which was never born, never changes, and never dies (KU 1.2.18; BG 2.20). This agrees with §1.3.3 that the body is not the self, but cuts deeper than joint form: the working mind and felt inner agent are also not-self.

3.3.3 The self is not individual; the question dissolves

The apparent separate person (*jiva*) is a provisional appearance caused by ignorance (*avidya*) — the superimposition (*adhyasa*) of self on not-self analysed in the preamble to Shankara’s *Brahma Sutra Bhashya* (BSB, *Adhyasa Bhashya*). Liberation (*moksha*) is not gaining anything new — it is recognising that the Self was always already Brahman: one, complete, without a second (CU 6.2.1, *tat tvam asi* at 6.8.7). “Is the human just material?” assumes a separate human in an independent material world. Advaita answers: ultimately there is no separate human and no independent material world; there is only Brahman, appearing as both — so the question **dissolves** rather than receiving a body-plus-soul answer.

3.4 Scripture, reason, and meditation

Advaita's warrant is not private realisation alone (contrast §1.4 and §2.4). The classical path is hearing the scriptures (*shravana*), reasoning over them (*manana*), and sustained meditation (*nididhyasana*), under a qualified teacher (BU 2.4.5). *Neti neti* is both phenomenological method (§3.2) and contemplative discipline: the final truth outruns positive language and is pointed to by negation (BU 2.3.6). Unlike Shri. Nagraj's **Realisation – Behaviour – Experiment** chain, which closes in conduct others can observe (§1.4), and unlike science's demand for inter-subjective test (§2.4), Advaita treats liberating insight as confirmed within a scriptural and contemplative lineage — a different epistemic standard, not a duplicate of public experiment.

3.5 What makes this answer distinctive

Among the three positions, Advaita makes the most radical anti-materialist claim. Consciousness is not emergent (§2.5) or a constitutionally complete atom (§1.3.2), but the ultimate reality in which both matter and mind appear. This purchases ontological parsimony at a cost: individual persons, relationships, and the world retain empirical force but not final independent reality. Madhyasth Darshan accepts the opposite cost — plural, enduring *jeevan* units and a perpetual world — in order to give individuality, relationship, and conduct full ontological weight.

4. Comparing the Three Views

Question	Physicalist interpretation	Madhyasth Darshan	Advaita Vedanta
What is a human?	An organism: body + brain + culture	Body + <i>jeevan</i> (a real sentient unit)	Ultimately <i>Atman</i> = <i>Brahman</i> ; the person is an appearance
Is the world real?	Yes; physical reality is ontologically sufficient	Yes — real and perpetual	Empirically operative but <i>mithya</i> , not ultimately independent
What is consciousness?	A capacity realised by organised brain–body activity	Active sentience in constitutionally complete <i>jeevan</i>	The ultimate reality in which matter and mind appear
How many selves?	Persons depend on individual organisms	Many <i>jeevan</i> units, forever distinct	Ultimately one Self, without a second
What survives death?	No person survives irreversible loss of brain organisation	<i>Jeevan</i> , intact, as a unit	<i>Atman</i> was never born; the separate person is not ultimate
Method of knowing	Public observation, modelling, experiment, replication	Study + self-verification + evidence in conduct	Scripture, reasoning, meditation under a teacher

Question	Physicalist interpretation	Madhyasth Darshan	Advaita Vedanta
Goal of life	No single scientific answer; naturalist ethics is a further inquiry	Awakening: resolved, humane living in society	Liberation: recognising the Self as Brahman

4.1 The key contrasts in plain words

On the body. Physicalism treats the embodied organism as ontologically sufficient. Advaita treats the body as empirically real but not ultimately independent. Shri. Nagraj holds that the body is real yet not the whole human: it is the necessary medium of an equally real *jeevan*. This comparative middle position is not the source of the philosophy’s name. *Madhyasth* names the mediative or regulative role through which Omnipresence and atomic nuclei sustain order ([The Ontology of Coexistence](#) §1.11). Nagraj’s counter-slogan to Vedanta preserves the world at the highest standpoint:

“Brahma is truth, the world is perpetual.”

— MVD, *The Alternative*, point 8

On the self. Physicalism finds no self over and above the embodied processes constituting a person. Advaita affirms one universal Self, with separate individuality sublated at the highest standpoint. Shri. Nagraj affirms many permanently distinct selves, giving relationships and values full ontological weight rather than only empirical validity.

On matter and mind. Physicalism treats mind as realised by matter’s organisation. Advaita does not derive matter from an individual or cosmic mind; it treats both matter and mind as appearances dependent on Brahman as consciousness. Madhyasth Darshan holds that atomic development reaches a sentient mode at constitutional completeness while refusing to reduce that mode to neural organisation.

On method. Science coordinates observation, instruments, first-person reports, models, experiments, and public error correction. Advaita joins scriptural testimony, reasoning, contemplative discrimination, and a teacher. Madhyasth Darshan joins study and first-person verification with evidence in conduct, work, and transmission. Each disputes whether the others’ evidence can distinguish truth from error: physicalism questions the ontological reach of contemplation and conduct, while both Indian views argue that third-person method does not exhaust the knower doing the inquiry.

Where the alliances fall. Against physicalism, Shri. Nagraj and Advaita agree that the human is not exhausted by bodily organisation. Against Advaita’s ultimate world-subordination, Shri. Nagraj and scientific realism agree that nature remains a genuine object of inquiry. Against both traditions, physicalism denies a persisting subject after brain death. Each pair therefore agrees on something the third denies without sharing a complete ontology.

Other anthropologies — Buddhist *anatta* (no permanent self), Abrahamic soul doctrines, and Dvaita-style eternal individual souls — offer further comparisons; see the wider table in [The Ontology of Coexistence](#) §5.

5. Critical Review

No view escapes serious objections. Here is where each is strong and where each is vulnerable.

5.1 Madhyasth Darshan — integrated proposal, unsettled ontology

Strengths. The account integrates consciousness, individuality, value, embodiment, and social responsibility in one real world. Unlike Advaita, it does not subordinate the world at the highest standpoint. Unlike constitutive panpsychism, it does not assemble a subject from billions of micro-subjects: one constitutionally complete unit is proposed as the bearer. That avoids the standard combination problem by restricting subjecthood, though it leaves the unity and sentience of the complete atom to be explained. Its evidence regime also refuses to stop at assertion or private experience: understanding must become visible in conduct, work, and transmission.

Weaknesses.

- **The foundational classification needs independent warrant.** The opening axiom distinguishes physical, chemical, and *jeevan* entities. The system is coherent if that classification is granted, but phenomenology and conduct do not independently establish the constitutionally complete unit. The issue is not that an argument may have premises; it is how this premise is justified to someone who does not already accept the ontology.
- **Different jobs do not prove two things.** The central argument — body serves food and sleep, *jeevan* serves values, therefore they are distinct entities — is a leap. One system can serve many purposes. And even granting distinctness, nothing in the phenomenological argument in §1.2 alone establishes that *jeevan* is *eternal*; that claim rests on the ontological argument in §1.3 and the conservation-based reading developed in [The Ontology of Coexistence](#).
- **Constitutional completeness is unexplained, not merely unmeasured.** No observation in particle physics or chemistry presently operationalises *gathanpurnata*. The deeper tension — whether sentience at completeness is latency actualised or an emergence the system otherwise resists — is analysed in [The Ontology of Coexistence](#) §§1.6.1, 6.2.1. Until the selection of this threshold is grounded, a sentient atom remains a metaphysical model rather than established physics.
- **The interaction relation is underspecified.** Conservation laws alone do not prove physical causal closure, but the joint-form account still needs to state how *jeevan* activity relates to neural activity and what observation would differ if *jeevan* were absent. The primary texts provide signal and medium language, not a publicly assessable mechanism ([The Ontology of Coexistence](#) §6.2.3).
- **Immortal individuality meets *anatta*.** Buddhist and Madhyamaka traditions deny an enduring self through aggregate decomposition or dependent origination — a pressure distinct from Advaita's witness but converging on post-death continuity ([The Ontology of Coexistence](#) §6.2.2).
- **Its validation can become circular.** A realised teacher and successful transmission support study, but teacher and doctrine can appear to validate one another. The tradition needs explicit

safeguards for disagreement, failed transmission, authority error, and cases in which sound conduct accompanies disputed metaphysics. Public assessment would require operational criteria for constitutional completeness, reproducible markers distinguishing *jeevan*-mediated from brain-only activity, and independent evidence for post-death continuity ([The Epistemology of Coexistence](#) §§7.2, 7.5–7.6).

Madhyasth Darshan is best read as a systematic and ethically rich proposal whose distinctive ontology remains to be established by evidence capable of distinguishing it from rivals.

5.2 Physicalism — powerful evidence, unfinished story

Strengths. Modern science has unmatched explanatory and predictive success in the domains it can investigate. Medicine, neuroscience, psychology, and evolutionary theory explain much that older systems assigned to souls. Public methods, controlled comparison, replication, and institutional criticism provide unusually powerful mechanisms for detecting empirical error.

Weaknesses.

- **The hard problem remains disputed.** Neural correlation and functional explanation do not by themselves explain why processing has a felt aspect. Some physicalists expect further mechanism or conceptual clarification to dissolve the gap; others accept that physicalism needs revision (§2.5). The gap is not evidence for *jeevan*, but it prevents functional success from closing the metaphysical question ([The Epistemology of Coexistence](#) §3.4).
- **First-person evidence is constrained but not eliminated.** Science can study reports, discrimination, attention, and behaviour, yet public operationalisation does not reproduce experience for an observer. The limitation is a coordination problem between first- and third-person evidence, not simple blindness to the knower.
- **Method is quietly promoted to metaphysics.** “Science detects only matter” is a fact about the method; “therefore only matter exists” is a philosophical leap. Many working scientists are methodological naturalists, not metaphysical materialists — the distinction is often blurred in debate.
- **It has little to say about meaning.** Values, purpose, and the question “how should we live?” are outside its scope. That is not a refutation, but it explains why purely materialist accounts of the human feel incomplete to many — the dissatisfaction Shri. Nagraj built his case on.

Scientific inquiry offers the strongest publicly corrected account of the body and its dependence relations. Physicalism’s claim to be the *complete* account of the human remains a philosophical interpretation of that success, not a scientific measurement.

5.3 Advaita Vedanta — profound, but at a steep price

Strengths. It takes the one undeniable datum — consciousness — as the starting point rather than an afterthought, which is philosophically defensible: experience is the only thing we know directly. It is supported by a sophisticated 1,200-year commentarial tradition, and its core insight (the observer cannot be any observed object) is a serious argument, not mere assertion.

Weaknesses.

- **The move from witness to Brahman exceeds phenomenology.** Seer–seen discrimination can challenge identification with body and thought, but it does not by itself establish one universal Self or the *mithya* status of the world. Those conclusions depend on wider reasoning and Upanishadic testimony.
- **It must explain dependent appearance.** If Brahman alone is ultimately real, what is the status of beginningless ignorance (*avidya*), whose is it, and how does plurality appear? Advaita’s sub-schools give different accounts. The world is not simply denied, but its dependence remains a substantive explanatory burden.
- **Ethics has empirical force but contested ultimate weight.** Advaita teaches disciplined conduct, non-injury, truthfulness, and responsibility within *vyavahara*. Madhyasth Darshan’s challenge is narrower: if individuality and relationship are ultimately sublated, can their significance have the same final ontological weight that its relational ethics requires?
- **The authority is scriptural.** The final warrant is the Upanishads as interpreted by the tradition. For anyone who does not grant scripture that status, the system’s deepest claims rest on testimony.

Advaita offers the most ontologically parsimonious answer of the three. Its cost is not that empirical questions become unanswerable, but that the persons and world about which they are answered do not retain final independent reality.

5.4 The bottom line

The disagreement concerns both **what exists** and **which evidence can discriminate rival accounts**. Science uses first-person reports but requires public control; Madhyasth Darshan requires coherence across realisation, conduct, work, and transmission; Advaita joins contemplative discrimination with scriptural testimony and reasoning. These standards are not arbitrary choices beyond comparison: each can be assessed for error correction, accessibility, predictive discrimination, practical consequence, and circularity, even when no single method settles every domain.

Different claims require different tests. A reproducible neural effect not explained by existing physical models would challenge current closure assumptions, but would not uniquely identify *jeevan*. Operational criteria for constitutional completeness and reproducible markers distinguishing *jeevan*-mediated from brain-only activity would bear directly on Madhyasth Darshan’s atomic and interaction claims. Independent evidence of post-death continuity would address persistence. A pre-registered conduct study could test whether Madhyasth Darshan practice produces distinctive resolution, relationship, and behavioural outcomes, but not whether those outcomes are caused by a sentient atom. A more complete mechanistic account of valuation would reduce one explanatory role assigned to *jeevan* without by itself resolving phenomenal consciousness or refuting Advaita’s *Atman* ([The Epistemology of Coexistence](#) §§7.2–7.4).

The three views are therefore intelligible within different starting commitments, but none has established every disputed step by evidence shared across all three. The study’s conclusion is not that the alternatives are equally supported. Modern science sets the strongest public constraints on body and

brain; Madhyasth Darshan offers an integrated but empirically underdetermined ontology of individual sentence; Advaita offers disciplined self-inquiry and ontological parsimony whose universal non-dual conclusion exceeds phenomenology alone.

Appendix: Quick Glossary

Key terms from §§1–5 are collected here for quick reference. Each term is also defined where it first appears in the exposition.

Term	Plain meaning
<i>Jeevan</i>	The sentient self — in Shri. Nagraj’s view a real, eternal, constitutionally complete unit that works <i>through</i> the body. It is neither brain activity nor an immaterial soul added from outside nature. For its ontology and conservation claims, see The Ontology of Coexistence §§1.6–1.8, 1.14, 6.2.
Constitutional completeness (<i>gathanpurnata</i>)	The claimed irreversible transition at which an evolving atom’s constitution closes and the unit becomes sentient <i>jeevan</i> . Developed in The Ontology of Coexistence §1.6.1.
Coexistence (<i>saha-astitva</i>)	Existence seen as all units inseparably present together.
Knowledge order (<i>gyan avastha</i>)	The human level of nature — able to know and evaluate every other level.
<i>Gyan udghatan</i>	The unfolding of knowledge in thought and conduct — an activity exclusive to awakened <i>jeevan</i> , not the brain or insentient matter (The Epistemology of Coexistence §1.1).
<i>Pramanikta</i>	The realisation-bearing authenticity that becomes evident in conduct, teaching, and humane tradition (The Epistemology of Coexistence §1.6).
<i>Rahasyata</i>	Mysteriousness — removable deficit in the knower’s reflection, not a permanent opacity of nature (The Epistemology of Coexistence §§3.4, 7.1).
Sensitivity / comprehension	Body- and sense-based responsiveness versus <i>jeevan</i> ’s cognisance of relationship, value, and order (The Epistemology of Coexistence §1.5).
Six perspectives (<i>drishti</i>)	Evaluative lenses in human behaviour — <i>priya/hita/labh</i> and humane <i>nyaya/dharma/satya</i> (The Ontology of Coexistence §1.12).
<i>Atman</i> / * Brahman *	In Advaita Vedanta: the innermost Self / the one ultimate reality. Advaita says they are identical.
<i>Mithya</i>	Advaita’s term for the world: not absolutely real, a dependent appearance.

References

Madhyasth Darshan (primary sources)

- **MVD** — Nagraj, A. [*Madhyasth Darshan – Co-existentialism, Part 1: Holistic View of Human Behaviour*](#). English translation by Rakesh Gupta. Cited: joint form (p. 13); *medhas* and activities of *jeevan* (p. 83); knowledge unfolding by awakened humans (pp. 115–116, 289); Ch. 3–4; *The Alternative* numbered points; constitutional completeness (p. 91).
- **SB** — Nagraj, A. [*Samadhanatmak Bhautikvad \(Resolution Centred Materialism\)*](#). English translation by Rakesh Gupta. Cited by chapter.
- **JV** — Nagraj, A. [*Jeevan Vidya: An Introduction*](#). English translation by Rakesh Gupta. Cited: values and evaluation not bodily processes (p. 39); human body provisioned to evidence understanding (p. 59); knowledge-order body requirement (pp. 79, 93); Ch. 1.

Related studies in this collection

- [*The Ontology of Coexistence*](#) — constitutional completeness, *jeevan* structure and harmonies, conservation, sentience latency, and *jeevan*–body interaction (§§1.6–1.8, 1.14, 6.2.1, 6.2.3).
- [*The Epistemology of Coexistence*](#) — evidence chain, active knower, *gyan udghatan*, joint-form knowing, sensitivity and comprehension, *rahasyata*, public testability, and physical causation (§§1.1, 1.3–1.6, 3.4, 7.1–7.4).
- [*Ethics-And-Morals-In-Human-Beings*](#) — comparative moral psychology and innate justice claims (§3.3, §6.1).

Advaita Vedanta (primary texts)

Verse and section numbers follow the standard numbering and apply to any faithful edition or translation (e.g. Swami Gambhirananda's or Swami Madhavananda's translations, Advaita Ashrama).

- **BU** — [*Brihadaranyaka Upanishad*](#). Cited: *neti neti* (2.3.6, 4.4.22); *shravana–manana–nididhyasana* (2.4.5).
- **TU** — [*Taittiriya Upanishad*](#). Cited: the five sheaths (*pancha-kosha*), 2.1–2.5.
- **MU** — [*Mandukya Upanishad*](#). Cited: the three states of consciousness (*avastha-traya*), vv. 3–7.
- **CU** — [*Chandogya Upanishad*](#). Cited: “one only, without a second” (6.2.1); *tat tvam asi* (6.8.7).
- **KU** — [*Katha Upanishad*](#). Cited: the unborn, undying Self (1.2.18).
- **BG** — [*Bhagavad Gita*](#). Cited: the indestructible self (2.20).
- **BSB** — Shankara, Adi. [*Brahma Sutra Bhashya*](#). Cited: the *Adhyasa Bhashya* (preamble) on superimposition of self and not-self.
- **VC** — [*Vivekachudamani*](#) (attributed to Shankara). Cited: discrimination between real and unreal (v. 20); sheath analysis (vv. 154–212); the deep-sleep witness (v. 120).

- **DDV** — [*Drig-Drishya-Viveka*](#) (attributed to Shankara or Bharati Tirtha). Cited: seer/seen discrimination (vv. 1–5).

Science and modern philosophy

- **Bloom 2013** — Bloom, P. [*Just Babies: The Origins of Good and Evil*](#). Crown.
- **Chalmers 1995** — Chalmers, D. “Facing Up to the Problem of Consciousness.” *Journal of Consciousness Studies*, 2(3), 200–219. Also at <https://consc.net/papers/facing.pdf>
- **Churchland 1986** — Churchland, P. S. [*Neurophilosophy: Toward a Unified Science of the Mind-Brain*](#). MIT Press.
- **Dennett 1991** — Dennett, D. [*Consciousness Explained*](#). Little, Brown.
- **Goff 2019** — Goff, P. [*Galileo’s Error: Foundations for a New Science of Consciousness*](#). Pantheon.
- **Kandel et al. 2021** — Kandel, E. R., et al. [*Principles of Neural Science*](#), 6th ed. McGraw-Hill. Cited for Ch. 39 (brain–machine interfaces).
- **Kim 2005** — Kim, J. [*Physicalism, or Something Near Enough*](#). Princeton University Press.
- **Nagel 1974** — Nagel, T. “What Is It Like to Be a Bat?” *The Philosophical Review*, 83(4), 435–450. Also at https://www.cs.ox.ac.uk/activities/ieg/e-library/sources/nagel_bat.pdf
- **Popper 1959** — Popper, K. [*The Logic of Scientific Discovery*](#). Hutchinson / Routledge.
- **Shapiro 2007** — Shapiro, L., ed. [*The Correspondence between Princess Elisabeth of Bohemia and René Descartes*](#). University of Chicago Press.
- **Strawson 2006** — Strawson, G. “Realistic Monism: Why Physicalism Entails Panpsychism.” *Journal of Consciousness Studies*, 13(10–11), 3–31. Also at <https://consc.net/event/reef/strawsonmonism.pdf>
- **Tomasello 2014** — Tomasello, M. [*A Natural History of Human Thinking*](#). Harvard University Press.